Converting Thesis & Capstone Projects to IMRaD

A Practical Guide for Condensing 5-Chapter Manuscripts into Publication-Ready Articles

Source Format: Traditional 5-Chapter Thesis / Capstone

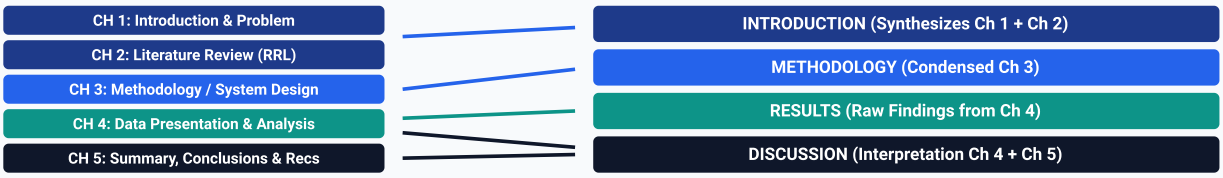
Target Format: Journal IMRaD Structure

Reduction Ratio: ~70% Content Distillation

Core Conversion Objective: Academic theses and capstones are written to demonstrate thoroughness to an academic committee (often 50–150 pages). In contrast, journal articles in **IMRaD format** (Introduction, Methodology, Results, and Discussion) communicate concise, novel findings to the scientific community (typically 10–25 pages).

1. THE 5-CHAPTER TO IMRAD MAPPING ARCHITECTURE

Converting a thesis or capstone requires shifting from an exhaustive academic record to a streamlined empirical narrative. The diagram below shows how traditional chapters consolidate into the four IMRaD blocks.



2. CONVERTING CHAPTERS 1 & 2 INTO THE INTRODUCTION

The thesis/capstone Chapters 1 and 2 often contain extensive operational definitions, local administrative background, and dozens of thematic literature summaries. In IMRaD, this must be compressed into a concise 3- to 5-page section.

- **Eliminate Generic Background:** Cut elementary textbook definitions and overly broad local descriptions (e.g., historical background of the university or region).
- **Synthesize the Literature (Ch 2):** Instead of listing study after study in standalone paragraphs, group citations by theme to highlight the specific *knowledge gap* or *engineering challenge*.
- **Integrate Frameworks & Objectives:** Convert Chapter 1's "Statement of the Problem" and "Conceptual Framework" into a smooth ending paragraph stating the exact research objective, hypothesis, or capstone system goal.

3. CONVERTING CHAPTER 3 INTO METHODOLOGY

Distilling Methodological Rigor without Thesis Verbosity

Chapter 3 of a thesis or capstone often includes step-by-step textbook explanations (e.g., explaining how Likert scales work or verbatim copying of software install scripts). IMRaD requires concise, reproducible descriptions of operational steps.

- **Research / System Architecture:** Clearly state the experimental design, sample size, or software architecture framework (e.g., Agile, Waterfall, ISO/IEC 25010) without defining standard software terms.
- **Data Collection & Instruments:** State what instruments were used, their validation metrics (Cronbach's alpha, hardware tolerances), and testing environments.
- **Data Analysis Plan:** Mention only the statistical tests, algorithms, or evaluation metrics actually executed in the study. Remove basic formula derivations unless novel.

4. CONVERTING CHAPTER 4 INTO RESULTS

Isolating Objective Findings from Subjective Interpretation

Traditional Chapter 4 sections combine raw data tables with extensive narrative explanations and subjective author commentary. For IMRaD, strip away all narrative interpretation and present pure empirical data.

- **Table & Figure Optimization:** Select only key summary tables and high-impact charts. Move secondary demographic tables or raw survey response logs to supplementary files.
- **Text Neutrality:** Report values, p-values, performance metrics, and accuracy percentages without using evaluative terms like "remarkably high" or "disappointingly low."

5. SECTION-BY-SECTION TRANSLATION MATRIX

Refer to this conversion table when determining where specific sub-sections of a traditional 5-chapter manuscript belong in IMRaD:

Traditional Thesis / Capstone Element	Target IMRaD Location	Action Required for Journal Formatting
Scope & Delimitation / Significance	**Introduction**	Incorporate scope into the research gap statement; remove standalone "Significance" subheadings.
Definition of Terms	**Methodology or Omit**	Delete standard dictionary definitions; define unique or technical terms inline upon first use.
System Architecture & Diagrams	**Methodology**	Include high-level design diagrams; remove step-by-step installation or wireframe setup screens.
Data Tables & Statistical Output	**Results**	Condense multi-page itemized survey tables into aggregated mean/standard deviation summary tables.
Interpretation of Data (Ch 4) + Ch 5	**Discussion**	Merge data discussion with Chapter 5 conclusions to explain findings against existing literature.

6. CONVERTING CHAPTERS 4 & 5 INTO DISCUSSION

The Discussion section is where thesis conversion requires the most restructuring. In a thesis, Chapter 4 interprets data while Chapter 5 provides summary, conclusions, and recommendations. In IMRaD, these elements are unified into a single coherent narrative.

- **Key Findings Summary:** Re-open with a direct statement answering the main research problem raised in Chapter 1.
- **Literary Comparison:** Connect your findings back to the Chapter 2 literature review. Highlight where your data agrees or disagrees with prior studies.
- **Thesis / Capstone Nuances:** For *Theses*, emphasize theoretical contributions and statistical significance. For *Capstones*, highlight system evaluation metrics, performance benchmarking, usability scores (e.g., SUS), and real-world deployment viability.
- **Limitations & Recommendations:** Replace Chapter 5's bulleted recommendations with a focused narrative discussing study limitations and strategic directions for future research.

7. CRITICAL CONTENT TRIMMING STRATEGY

✂ Content to Delete Completely

Elementary textbook definitions, basic code snippets, raw item-by-item survey logs, committee approval sheets, acknowledgments (move to endnote), and long organizational histories.

✓ Content to Keep & Refine

Core research questions, sampling methods, validated instruments, summary data tables, statistical significance outputs, and direct answers to hypotheses/objectives.

🔬 Empirical Thesis Focus

Emphasize conceptual frameworks, variable relationships, statistical power, error rates, and theoretical implications within the broader scientific literature.

💻 Capstone Project Focus

Emphasize system requirements, architecture flow, functional evaluation, ISO/IEC quality metrics, user acceptance testing, and practical deployment feasibility.

8. PRE-SUBMISSION QUALITY CHECKLIST

Author Checklist: Thesis/Capstone to IMRaD Readiness

- ✓ **Abstract Rewritten:** Compressed to 150–250 words following a 4-part mini-IMRaD structure.
- ✓ **Chapter Headings Removed:** Replaced "Chapter 1", "Chapter 2", etc., with standard IMRaD headers.
- ✓ **Word Count Target Achieved:** Reduced original manuscript by 60–80% down to target journal length.
- ✓ **References Consolidated:** Pruned bibliography to retain only directly cited, peer-reviewed sources.